

Spiral Similarities and Ratios

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NOTE: This is an adaptation of my handout with the same name which I used in the Canada 2022 Winter Camp. Also note that problem sources are not given in text for psychological reasons, but are given in the LaTeX file in comments.

Olympiad geometry is often taught like this: you're given a problem with a bunch of random components. You don't know how to deal with the components. Well, what do we do? It turns out that just through some chance, all these random components fit together with this ONE similarity and this ONE pair of ratios that just solve all the angles and make everything nice. And before you can get to ask how anyone would ever think of comparing those to triangles, you're asked to solve a completely different problem with a completely different diagram.

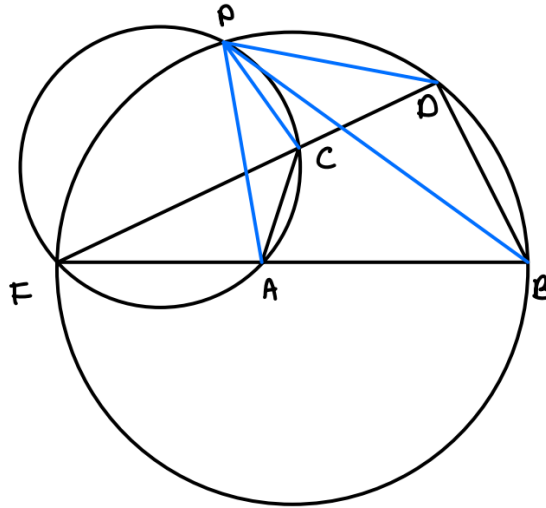
It's hopeless, right?

Today's lecture on spiral similarities is going to teach you that it's **not**. In fact, these similar triangles, 99% of the time, come from - you guessed it - spiral similarities.

1 Main Lecture

1.1 Introducing Spiral Similarities

Given any four distinct points A, B, C, D , there exists a **unique** point P such that the triangle PAB is directly similar to PCD . We hence call P the *spiral center* taking AB to CD . To construct this point, consider $AB \cap CD = F$. Note that $\angle PDC = \angle PBA$ is equivalent to $P \in (FBD)$, and similarly we must also have $P \in (FCA)$. Hence the intersection of these two circles is the desired point P .



A nice property of this point P is that this spiral similarity is not isolated; specifically, we have that if P

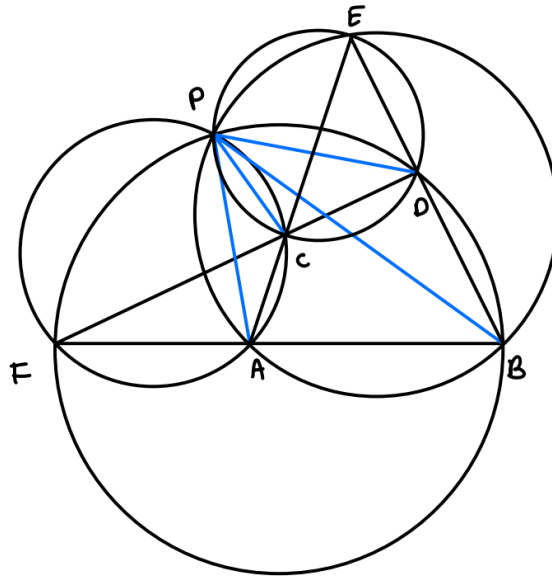
is the spiral center taking AB to CD , then it also takes AC to BD . This is because since

$$\begin{aligned}\angle APC &= \angle BPC - \angle APB \\ &= \angle BPC - \angle CPD \\ &= \angle BPD\end{aligned}$$

and

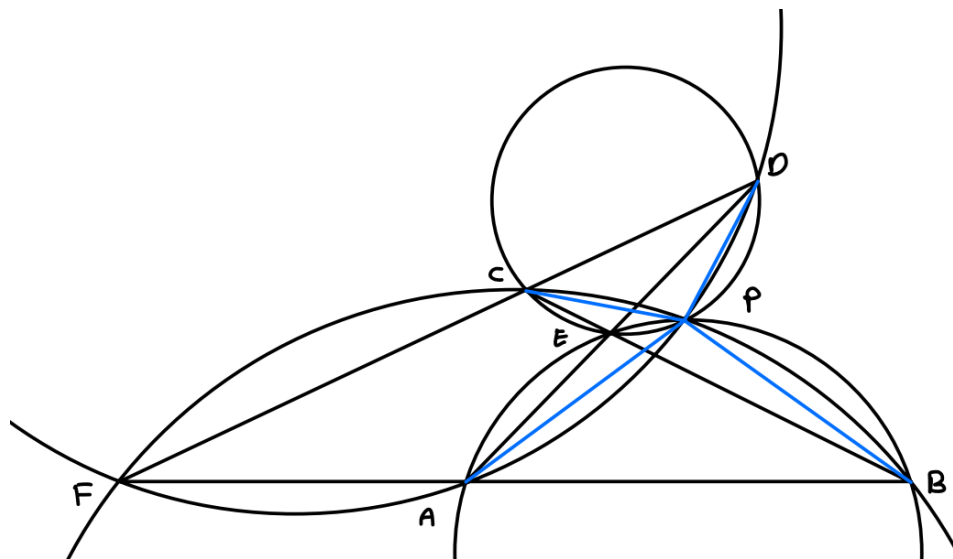
$$\begin{aligned}\frac{AP}{BP} &= \frac{CP}{DP} \\ \frac{AP}{CP} &= \frac{BP}{DP}\end{aligned}$$

we have that $PAC \sim PBD$, as desired.



This is useful for two reasons. One, from one pair of similar triangles we have found another that came automatically! Moreover, our construction of P still applies to this other pair: if $AC \cap BD = E$, then $P = (ABE) \cap (CDE)$. So in fact this implies that there are four circles here - (ABE) , (CDE) , (ACF) , and (BDF) - and they all share the common point P , which is also the spiral center. P is hence also known as the *Miquel point* of $ABDC$.

Remark 1.1 *There is some depth in the notation here. Given a quadrilateral $WXYZ$, the corresponding complete quadrilateral is these four points W, X, Y, Z in order, along with the intersections $WX \cap YZ$ and $XY \cap ZW$. Using the procedure above, we may correspond to any complete quadrilateral a Miquel point. But if we were to interchange the order of the vertices, we would obtain a different complete quadrilateral, and hence a distinct Miquel point! See the diagram below for a demonstration of what the configuration above would be if we instead considered the complete quadrilateral formed by $ABCD$.*



Here is a common scenario in which spiral similarities show up.

Example 1.2 Let ABC be a triangle, and let $D \in BC$. Let O_1 and O_2 be the circumcenters of ABD and ABC , respectively. Then prove that $AO_1O_2 \sim ADC$.

As an intermediate hint: try to prove $AO_1D \sim AO_2C$. What kinds of triangles are both of these?

Here is a nice problem demonstrating the usefulness of spiral similarities.

Example 1.3 Circles ω_1 and ω_2 meet at P and Q . Segments AC and BD are chords of ω_1 and ω_2 respectively, such that segment AB and ray CD meet at P . Ray BD and segment AC meet at X . Point Y lies on ω_1 such that $PY \parallel BD$. Point Z lies on ω_2 such that $PZ \parallel AC$. Prove that points Q, X, Y, Z are collinear.

Walkthrough:

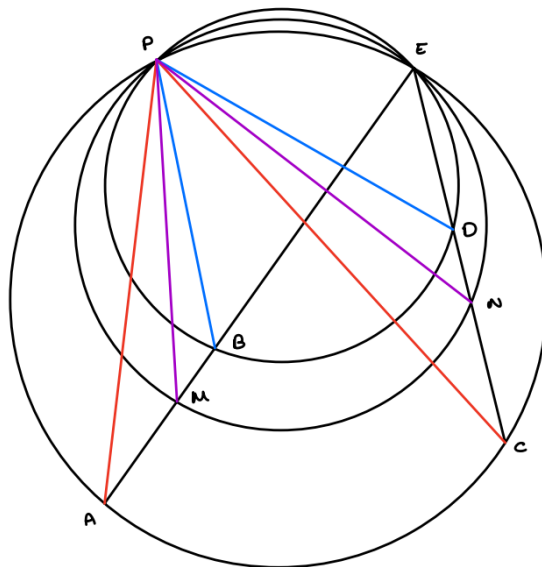
1. Consider the quadrilateral $APDX$. What is Q called?
2. What circles do Q lie on as a result?
3. Using this, prove (by angle chasing or otherwise) that X , Q , and Z are collinear, and X , Q , and Y are collinear. This hence proves the problem.

1.1.1 Exercises

1. Consider an arbitrary quadrilateral $ABCD$ such that $AB \cap CD = P$, $BC \cap DA = Q$. Let the circumcenters of PBC , PDA , QAB , and QCD be O_1 , O_2 , O_3 , and O_4 , respectively. Prove that $O_1O_2O_3O_4$ is cyclic, and the circumcircle of this cyclic quadrilateral passes through the Miquel point of $ABCD$.
2. Let ABC be a triangle, and let D be a point on side BC . A line through D intersects side AB at X and ray AC at Y . The circumcircle of triangle BXD intersects the circumcircle ω of triangle ABC again at point Z distinct from point B . The lines ZD and ZY intersect ω again at V and W respectively. Prove that $AB = VW$.
3. Circles S_1 and S_2 intersect at points P and Q . Distinct points A_1 and B_1 (not at P or Q) are selected on S_1 . The lines A_1P and B_1P meet S_2 again at A_2 and B_2 respectively, and the lines A_1B_1 and A_2B_2 meet at C . Prove that, as A_1 and B_1 vary, the circumcentres of triangles A_1A_2C all lie on one fixed circle.

1.2 Ratios

One very useful property about spiral similarities is that if $PAB \sim PCD$, and M and N are points on AB and CD respectively such that $\frac{AM}{MB} = \frac{CN}{ND}$, then the spiral similarity centered at P taking AB to CD also takes M to N . In other words, we also have that $PAM \sim PCN$ and $PMB \sim PND$.



This property is especially useful because it really allows us to get a lot of difficult-to-reach angles from spiral similarities. Midpoints and the like are often very annoying to deal with - with this condition however, this allows us to work with their angles just like usual.

Here's a classic example:

Example 1.4 Let $ABCD$ be a quadrilateral, and let E and F be points on sides AD and BC , respectively, such that $\frac{AE}{ED} = \frac{BF}{FC}$. Ray FE meets rays BA and CD at S and T , respectively. Prove that the circumcircles of triangles SAE , SBF , TCF , and TDE pass through a common point.

Walkthrough:

1. What is $(SAE) \cap (SBF)$ called?
2. What is $(TED) \cap (TFC)$ called?
3. Why are these the same point?

In particular, what this demonstrates is that spiral similarities work really well with **ratios**. In particular, since every pair of directly similar triangles gives you a new similarity, and these pairs can *themselves* be derived from previous ratios, this implies that often chasing ratios that may seem to be useful may work very well in tandem with spiral similarities.

Example 1.5 Let ABC be a triangle and let D , E , F be the projections of the incenter of ABC onto BC , AC , AB respectively. Let the circumcircles of ABC and AEF intersect at K . Prove that KD bisects $\angle BKC$.

Walkthrough:

1. In terms of the angle bisector theorem, what would we need to prove in order to show that KD bisects $\angle BKC$?
2. K is a Miquel point. Identify what quadrilateral it is the Miquel point of, and give a few ratios that result from this spiral similarity.

3. Tie the ratios together to prove the claim.

Example 1.6 Triangle ABC is inscribed in circle Ω . The interior angle bisector of angle A intersects side BC and Ω at D and L (other than A), respectively. Let M be the midpoint of side BC . The circumcircle of triangle ADM intersects sides AB and AC again at Q and P (other than A), respectively. Let N be the midpoint of segment PQ , and let H be the foot of the perpendicular from L to line ND . Prove that line ML is tangent to the circumcircle of triangle HMN .

Walkthrough:

1. Prove that the problem is equivalent to showing that $AL \parallel MN$. This gets rid of H (the most mysterious point in the diagram anyway).
2. Let's now actually work with the problem. What does $(AMD) \cap (ABC)$ seem to be? Call this point S , and prove your proposition. You should have that $S \in ML$.
3. Consider the quadrilateral $BCPQ$. What is S in relation to this quadrilateral?
4. What triangles are SNM similar to?
5. Using this, prove that $\angle SMN = \angle SLA$, and use this to finish the problem.

Example 1.7 Let $\triangle ABC$ be an acute-angled triangle with $AB \neq AC$. Let H be the orthocenter of triangle ABC , and let M be the midpoint of the side BC . Let D be a point on the side AB and E a point on the side AC such that $AE = AD$ and the points D, H, E are on the same line. Prove that the line HM is perpendicular to the common chord of the circumscribed circles of triangle $\triangle ABC$ and triangle $\triangle ADE$.

Walkthrough:

1. First, some setup. Let $Y = BH \cap AC$ and $Z = CH \cap AB$. Prove that if $K = (AYZ) \cap (ABC)$ and $K \neq A$, then K also lies on HM .
2. The main idea here is to prove that $AEKD$ is cyclic. Using spiral similarities, prove that this is equivalent to proving that

$$\frac{ZD}{DB} = \frac{YE}{EC}$$

3. Prove this ratio. It may help to recall that $ZC \cap YB = H$.

In a sense, what this implies is that a good way of thinking about these spiral similarities is as a sort of **algebraic manipulation**: by manipulating these ratios and spiral similarities, we can "reach" a lot of angles that would be previously unobtainable! In particular, some of the harder examples will demonstrate that this approach of algebraic manipulation to push for the exact similarities you want is exactly what is needed to solve many problems.

1.2.1 Exercises

1. In triangle ABC , points P, Q, R lie on sides BC, CA, AB respectively. Let $\omega_A, \omega_B, \omega_C$ denote the circumcircles of triangles AQR, BRP, CPQ , respectively. Given the fact that segment AP intersects $\omega_A, \omega_B, \omega_C$ again at X, Y, Z , respectively, prove that $YX/XZ = BP/PC$.
2. Let $ABCDE$ be a convex pentagon such that

$$\angle BAC = \angle CAD = \angle DAE \quad \text{and} \quad \angle ABC = \angle ACD = \angle ADE.$$

The diagonals BD and CE meet at P . Prove that the line AP bisects the side CD .

3. Let $ABCD$ be a fixed convex quadrilateral with $BC = DA$ and BC not parallel with DA . Let two variable points E and F lie of the sides BC and DA , respectively and satisfy $BE = DF$. The lines AC and BD meet at P , the lines BD and EF meet at Q , the lines EF and AC meet at R .

Prove that the circumcircles of the triangles PQR , as E and F vary, have a common point other than P .

1.3 Similar Figures

Not only are the above results true for ratios, but they also work with similar figures! What I mean is this: if we have the spiral center P taking AB to CD , and E and F are points such that ABE and CDF themselves are directly similar, then this spiral similarity also takes E to F (as the two triangle similarities imply that $PABE \sim PCDF$.)

This allows us to tie together figures that are significantly more "arbitrary" into a similarity as well! A key example:

Example 1.8 In an acute triangle ABC the points D, E and F are the feet of the altitudes through A, B and C respectively. The incenters of the triangles AEF and BDF are I_1 and I_2 respectively; the circumcenters of the triangles ACI_1 and BCI_2 are O_1 and O_2 respectively. Prove that I_1I_2 and O_1O_2 are parallel.

Walkthrough:

1. Draw a good diagram :)
2. First off; there's no reason not to add the third incenter I_3 of CDE , so draw that in too. What circles do this incenter lie on (don't prove anything yet)?
3. Let's now prove that AI_1I_2B is cyclic. Prove that this is equivalent to proving that $I_1I_2F \sim ADF$.
4. Let's flip the spiral sim, and prove that $AI_1F \sim DI_2F$. Prove that $AEF \sim DBF$, and demonstrate that this implies $AI_1F \sim DI_2F$.
5. With the key claim proved, let's now move on to the problem. Prove that it suffices to show $CI \perp I_1I_2$, where I is the incenter of the triangle.
6. Prove this by angle chasing or otherwise.
7. Where did we use the similar figures argument?

1.4 Review

1. The **Miquel point** P of a quadrilateral $ABCD$ is the spiral center taking AB to DC and AD to BC . If $AB \cap CD = E$ and $AD \cap BC = F$, then P lies on (ADE) , (BCE) , (ABF) , and (CDF) .
2. If $M, N \in AB, CD$ respectively such that $\frac{AM}{MB} = \frac{DN}{NC}$, then we have that P also takes AMB to DMC . This property allows us to deal with the angles of points that aren't as tied-into the diagram as the other points.
 - (a) The above also applies for points M, N satisfying that $AMB \sim DMC$.
3. To get the most out of spiral similarities, it can often help to often ratio-chase to really eek out the most information from the ratio property.

2 Problems

Here are some general problems regarding the materials of this handout that are less specific than the section exercises above. I've split the problems into three parts: Part A (IMO 1 or easier), Part B (IMO 2), and Part C (IMO 3). ★s represent problems I think are particularly good illustrations of the themes of this lecture.

2.1 Part A

1. Let $\triangle ABC$ be an acute-angled triangle, and let D be the foot of the altitude from C . The angle bisector of $\angle ABC$ intersects CD at E and meets the circumcircle ω of triangle $\triangle ADE$ again at F . If $\angle ADF = 45^\circ$, show that CF is tangent to ω .
2. ★ Let $ABCD$ be a cyclic quadrilateral. Let P, Q, R be the feet of the perpendiculars from D to the lines BC, CA, AB , respectively. Show that $PQ = QR$ if and only if the bisectors of $\angle ABC$ and $\angle ADC$ are concurrent with AC .
3. We say that a triangle ABC is great if the following holds: for any point D on the side BC , if P and Q are the feet of the perpendiculars from D to the lines AB and AC , respectively, then the reflection of D in the line PQ lies on the circumcircle of the triangle ABC . Prove that triangle ABC is great if and only if $\angle A = 90^\circ$ and $AB = AC$.
4. ★ Let O denote the circumcentre of an acute-angled triangle ABC . Let point P on side AB be such that $\angle BOP = \angle ABC$, and let point Q on side AC be such that $\angle COQ = \angle ACB$. Prove that the reflection of BC in the line PQ is tangent to the circumcircle of triangle APQ .
5. Let ABC be a fixed acute triangle inscribed in a circle ω with center O . A variable point X is chosen on minor arc AB of ω , and segments CX and AB meet at D . Denote by O_1 and O_2 the circumcenters of triangles ADX and BDX , respectively. Determine all points X for which the area of triangle OO_1O_2 is minimized.
6. Let ABC be a triangle with $AB < AC$. Let D be the intersection point of the internal bisector of angle BAC and the circumcircle of ABC . Let Z be the intersection point of the perpendicular bisector of AC with the external bisector of angle $\angle BAC$. Prove that the midpoint of the segment AB lies on the circumcircle of triangle ADZ .
7. Let ABC be a triangle such that $AB > BC$ and let D be a variable point on the line segment BC . Let E be the point on the circumcircle of triangle ABC , lying on the opposite side of BC from A such that $\angle BAE = \angle DAC$. Let I be the incenter of triangle ABD and let J be the incenter of triangle ACE . Prove that the line IJ passes through a fixed point, that is independent of D .
8. ★ Let $ABCD$ be a parallelogram. Points P and Q lie inside $ABCD$ such that $\triangle ABP$ and $\triangle BCQ$ are equilateral. Prove that the intersection of the line through P perpendicular to PD and the line through Q perpendicular to DQ lies on the altitude from B in $\triangle ABC$.

2.2 Part B

1. ★ The isosceles triangle $\triangle ABC$, with $AB = AC$, is inscribed in the circle ω . Let P be a variable point on the arc $\widehat{BC}$ that does not contain A , and let I_B and I_C denote the incenters of triangles $\triangle ABP$ and $\triangle ACP$, respectively.
Prove that as P varies, the circumcircle of triangle $\triangle PI_BI_C$ passes through a fixed point.
2. Let ABC be a triangle with $\angle C = 90^\circ$, and let H be the foot of the altitude from C . A point D is chosen inside the triangle CBH so that CH bisects AD . Let P be the intersection point of the lines BD and CH . Let ω be the semicircle with diameter BD that meets the segment CB at an interior point. A line through P is tangent to ω at Q . Prove that the lines CQ and AD meet on ω .

3. ★ Let O be the circumcenter of an acute triangle ABC . Line OA intersects the altitudes of ABC through B and C at P and Q , respectively. The altitudes meet at H . Prove that the circumcenter of triangle PQH lies on a median of triangle ABC .
4. Let $ABCD$ be an isosceles trapezoid with $AB \parallel CD$. Let E be the midpoint of AC . Denote by ω and Ω the circumcircles of the triangles ABE and CDE , respectively. Let P be the crossing point of the tangent to ω at A with the tangent to Ω at D . Prove that PE is tangent to Ω .
5. ★ Points A_1, B_1, C_1 are chosen on the sides BC, CA, AB of a triangle ABC respectively. The circumcircles of triangles $AB_1C_1, BC_1A_1, CA_1B_1$ intersect the circumcircle of triangle ABC again at points A_2, B_2, C_2 respectively ($A_2 \neq A, B_2 \neq B, C_2 \neq C$). Points A_3, B_3, C_3 are symmetric to A_1, B_1, C_1 with respect to the midpoints of the sides BC, CA, AB respectively. Prove that the triangles $A_2B_2C_2$ and $A_3B_3C_3$ are similar.
6. Let ABC be a triangle and let M and N denote the midpoints of $\overline{AB}$ and $\overline{AC}$, respectively. Let X be a point such that $\overline{AX}$ is tangent to the circumcircle of triangle ABC . Denote by ω_B the circle through M and B tangent to $\overline{MX}$, and by ω_C the circle through N and C tangent to $\overline{NX}$. Show that ω_B and ω_C intersect on line BC .
7. Let $ABCD$ be a quadrilateral with $AC = BD$. Diagonals AC and BD meet at P . Let ω_1 and O_1 denote the circumcircle and the circumcenter of triangle ABP . Let ω_2 and O_2 denote the circumcircle and circumcenter of triangle CDP . Segment BC meets ω_1 and ω_2 again at S and T (other than B and C), respectively. Let M and N be the midpoints of minor arcs $\widehat{SP}$ (not including B) and $\widehat{TP}$ (not including C). Prove that $MN \parallel O_1O_2$.
8. ★ Given a triangle ABC with $AB > BC$, let Ω be the circumcircle. Let M, N lie on the sides AB, BC respectively, such that $AM = CN$. Let K be the intersection of MN and AC . Let P be the incentre of the triangle AMK and Q be the K -excentre of the triangle CNK . If R is midpoint of the arc ABC of Ω then prove that $RP = RQ$.
9. ★ A convex quadrilateral $ABCD$ has perpendicular diagonals. The perpendicular bisectors of the sides AB and CD meet at a unique point P inside $ABCD$. Prove that the quadrilateral $ABCD$ is cyclic if and only if triangles ABP and CDP have equal areas. (solve this synthetically!)
10. Oscar is drawing diagrams with trash can lids and sticks. He draws a triangle ABC and a point D such that DB and DC are tangent to the circumcircle of ABC . Let B' be the reflection of B over AC and C' be the reflection of C over AB . If O is the circumcenter of $DB'C'$, help Oscar prove that AO is perpendicular to BC .
11. Let ABC be an acute-angled triangle with circumcircle ω and circumcentre O . Points $D \neq B$ and $E \neq C$ lie on ω such that $BD \perp AC$ and $CE \perp AB$. Let CO meet AB at X , and BO meet AC at Y . Prove that the circumcircles of triangles BXD and CYE have an intersection lie on line AO .
12. Let ABC be a triangle with circumcircle Γ and incenter I and let M be the midpoint of $\overline{BC}$. The points D, E, F are selected on sides $\overline{BC}, \overline{CA}, \overline{AB}$ such that $\overline{ID} \perp \overline{BC}$, $\overline{IE} \perp \overline{AI}$, and $\overline{IF} \perp \overline{AI}$. Suppose that the circumcircle of $\triangle AEF$ intersects Γ at a point X other than A . Prove that lines XD and AM meet on Γ .
13. ★ Circles W_1, W_2 intersect at P, K . XY is common tangent of two circles which is nearer to P and X is on W_1 and Y is on W_2 . XP intersects W_2 for the second time in C and YP intersects W_1 in B . Let A be intersection point of BX and CY . Prove that if Q is the second intersection point of circumcircles of ABC and AXY

$$\angle QXA = \angle QKP$$

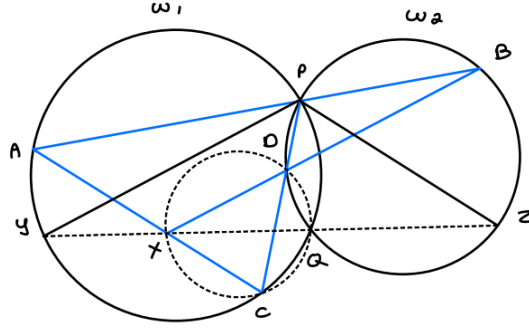
2.3 Part C

1. Let P be a point on the circumcircle of acute triangle ABC . Let D, E, F be the reflections of P in the A -midline, B -midline, and C -midline. Let ω be the circumcircle of the triangle formed by the perpendicular bisectors of AD, BE, CF .
Show that the circumcircles of $\triangle ADP, \triangle BEP, \triangle CFP$, and ω share a common point.
2. Let ABC be a triangle with $AB < AC$, incenter I , and A excenter I_A . The incircle meets BC at D . Define $E = AD \cap BI_A$, $F = AD \cap CI_A$. Show that the circumcircle of $\triangle AID$ and $\triangle I_AEF$ are tangent to each other.
3. ★ Let ω be the incircle of the triangle ABC and with centre I . Let Γ be the circumcircle of the triangle AIB . Circles ω and Γ intersect at the point X and Y . Let Z be the intersection of the common tangents of the circles ω and Γ . Show that the circumcircle of the triangle XYZ is tangent to the circumcircle of the triangle ABC .
4. ★ An equilateral pentagon $AMNPQ$ is inscribed in triangle ABC such that $M \in \overline{AB}$, $Q \in \overline{AC}$, and $N, P \in \overline{BC}$. Let S be the intersection of $\overrightarrow{MN}$ and $\overrightarrow{PQ}$. Denote by ℓ the angle bisector of $\angle MSQ$.
Prove that $\overline{OI}$ is parallel to ℓ , where O is the circumcenter of triangle ABC , and I is the incenter of triangle ABC .
5. ★ Let $ABCD$ be a non-cyclical, convex quadrilateral, with no parallel sides. The lines AB and CD meet in E . Let $M \neq E$ be the intersection of circumcircles of ADE and BCE . The internal angle bisectors of $ABCD$ form an convex, cyclical quadrilateral with circumcenter I . The external angle bisectors of $ABCD$ form an convex, cyclical quadrilateral with circumcenter J .
Show that I, J, M are collinear.
6. ★ Let ABC be an acute triangle with circumcenter O and orthocenter H . Let Γ denote the circumcircle of triangle ABC , and N the midpoint of OH . The tangents to Γ at B and C , and the line through H perpendicular to line AN , determine a triangle whose circumcircle we denote by ω_A . Define ω_B and ω_C similarly. Prove that the common chords of ω_A, ω_B and ω_C are concurrent on line OH .

3 Solutions

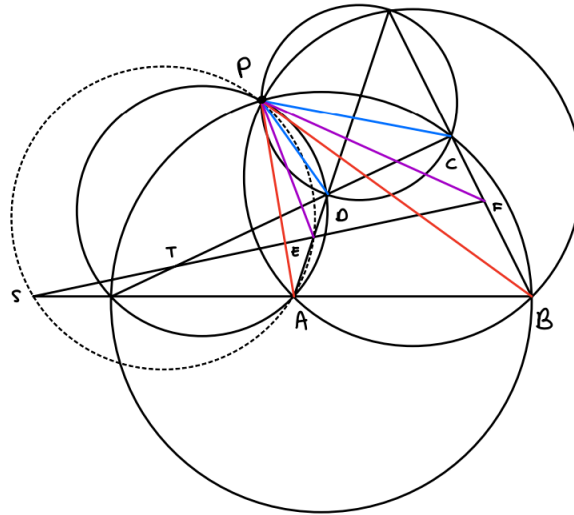
Example 3.1 Circles ω_1 and ω_2 meet at P and Q . Segments AC and BD are chords of ω_1 and ω_2 respectively, such that segment AB and ray CD meet at P . Ray BD and segment AC meet at X . Point Y lies on ω_1 such that $PY \parallel BD$. Point Z lies on ω_2 such that $PZ \parallel AC$. Prove that points Q, X, Y, Z are collinear.

Note that Q is the Miquel point of $APDX$. This implies that $QDXC$ is a cyclic quadrilateral. Now the parallel condition gives us that $\angle DQX = \angle DCX = \angle DPZ = 180^\circ - \angle DQZ$, so X, Q , and Z are collinear. Similarly, since $\angle YQC = \angle YPC = \angle XDC = \angle XQC$, we have that Y, X , and Q are collinear as well, thus solving the problem.



Example 3.2 Let $ABCD$ be a quadrilateral, and let E and F be points on sides AD and BC , respectively, such that $\frac{AE}{ED} = \frac{BF}{FC}$. Ray FE meets rays BA and CD at S and T , respectively. Prove that the circumcircles of triangles SAE , SBF , TCF , and TDE pass through a common point.

Consider the spiral center K taking AE to BF . Since $\frac{AE}{ED} = \frac{BF}{FC}$, we have that this spiral center also takes ED to FC . Hence K is the Miquel point of both $ABFE$ and $EFCD$, and so $K \in (SAE), (SBF), (TED)$, and (TFC) , thus solving the problem.

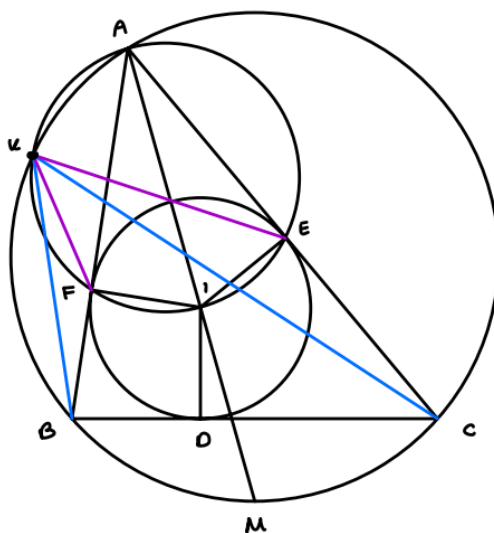


Example 3.3 Let ABC be a triangle and let D, E, F be the projections of the incenter of ABC onto BC, AC, AB respectively. Let the circumcircles of ABC and AEF intersect at K . Prove that KD bisects $\angle BKC$.

Note that K is the spiral center taking FE to BC . This hence implies that:

$$\begin{aligned} \frac{KB}{KC} &= \frac{FB}{EC} \\ &= \frac{DB}{DC} \end{aligned}$$

with the last equality arising from the fact that $FB = DB$ and $EC = DC$ by incircle properties. This thus implies that DK bisects $\angle BKC$, as desired.

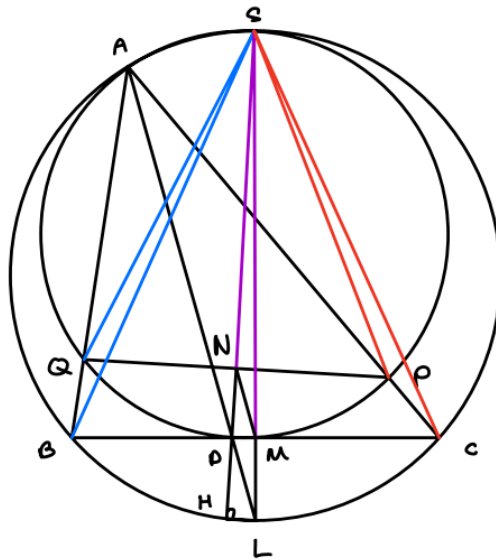


Example 3.4 Triangle ABC is inscribed in circle Ω . The interior angle bisector of angle A intersects side BC and Ω at D and L (other than A), respectively. Let M be the midpoint of side BC . The circumcircle of triangle ADM intersects sides AB and AC again at Q and P (other than A), respectively. Let N be the midpoint of segment PQ , and let H be the foot of the perpendicular from L to line ND . Prove that line ML is tangent to the circumcircle of triangle HMN .

First, note the tangency is equivalent to $\angle NML = 180^\circ - \angle NMH = 180^\circ - \angle DLM$ (where we use that $DMLH$ is cyclic). So it suffices to prove NM and AL are parallel.

Now, let S be the midpoint of arc BAC . Note that $\angle SAD = \angle DML = 90^\circ$, so $SADM$ is cyclic. Also, SL is the perpendicular bisector of BC , so S, M, L are collinear. Notably S is the Miquel point of $BCPQ$. Thus, since N and M are the midpoints of PQ and CB respectively, we have that $SPNQ \sim SCMB$.

Now we can finish. Note that we thus have $SNM \sim SPC$, so $\angle SMN = \angle SCP = \angle SLA$, hence AL and NM are parallel, as desired.



Example 3.5 Let $\triangle ABC$ be an acute-angled triangle with $AB \neq AC$. Let H be the orthocenter of triangle ABC , and let M be the midpoint of the side BC . Let D be a point on the side AB and E a point on the side AC such that $AE = AD$ and the points D, H, E are on the same line. Prove that the line HM is perpendicular to the common chord of the circumscribed circles of triangle $\triangle ABC$ and triangle $\triangle ADE$.

Let $Y = BH \cap AC$, $Z = CH \cap AB$, and A' be the antipode of A on (ABC) . note that $BHCA'$ is a parallelogram (both BA' and CH are perpendicular to AB), so A' lies on HM . But now if $K = A'H \cap (ABC)$, then we know that $\angle AKH = \angle AK A' = 90^\circ$. Hence K lies on (AH) , and so $K \in (AYZ)$. Thus, K is the spiral center taking ZY to BC .

Now, note that $\angle ADE = 90^\circ - \frac{1}{2}\angle A = \frac{1}{2}(90^\circ + 90^\circ - \angle A) = \frac{1}{2}(\angle AZC + \angle ABY)$. This thus implies that $\angle DHB = \frac{1}{2}\angle ZHB$, so thus DE bisects $\angle ZHB$. Hence, we have that by the Angle Bisector theorem,

$$\frac{ZD}{DB} = \frac{ZH}{HB}$$

$$\frac{YE}{EC} = \frac{YH}{HC}$$

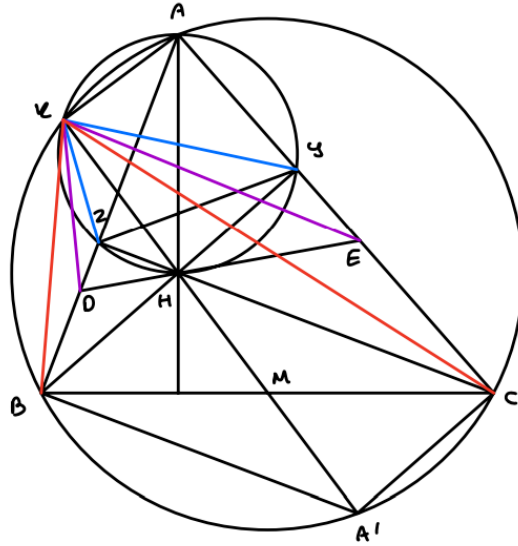
But in fact $\angle HZB = \angle HYC = 90^\circ$, and $\angle HBZ = \angle HCY$ (as $BZYC$ is cyclic), so thus $HBZ \sim HCY$. This thus implies that:

$$\frac{ZH}{HB} = \frac{YH}{HC}$$

so thus:

$$\frac{ZD}{DB} = \frac{YE}{EC}$$

But now consider K ; we have that since K takes BZ to CY through a spiral sim, it also takes BD to CE . But then K is the Miquel point of $BDEC$, so since $BD \cap CE = A$ we have that $K \in (ADE)$. Since $K \in HM, (ABC)$ as well, this thus solves the problem!



Example 3.6 In an acute triangle ABC the points D, E and F are the feet of the altitudes through A, B and C respectively. The incenters of the triangles AEF and BDF are I_1 and I_2 respectively; the circumcenters of the triangles ACI_1 and BCI_2 are O_1 and O_2 respectively. Prove that I_1I_2 and O_1O_2 are parallel.

The main claim is that AI_1I_2B is cyclic. To prove this, note first that $AEF \sim DBF$ by angle chasing (in particular, this is a spiral similarity). Hence this implies that F takes the figure FAE to FDB , and in this spiral similarity also takes I_1 to I_2 . So $FI_1I_2 \sim FAD$. Now this implies that:

$$\begin{aligned} \angle I_1AB + \angle I_1I_2B &= \frac{1}{2} \angle A + \angle ADF + \angle FI_2B \\ &= \frac{1}{2} \angle A + \angle ACF + 90^\circ + \frac{1}{2} \angle FDB \\ &= \frac{1}{2} \angle A + 90^\circ - \angle A + 90^\circ + \frac{1}{2} \angle A \\ &= 180^\circ \end{aligned}$$

which gives us that AI_1I_2B is cyclic, as desired. ■

Now, let I_3 be the incenter of CDE , and I the incenter of ABC . Note that AI_1 , BI_2 , and CI_3 all pass through CI . In addition, by an analogous proof to the one above, we have that BI_2I_3C and CI_3I_1A are both cyclic as well. this implies that $O_1O_2 \perp CI_3$. However, note that:

$$\begin{aligned} \angle CIA &= 90^\circ + \frac{1}{2} \angle B \\ &= 90^\circ + \angle IBA \\ &= 90^\circ + \angle I I_1 I_2 \end{aligned}$$

This thus implies that $I_1I_2 \perp CI$ as well, and hence $I_1I_2 \parallel O_1O_2$, as desired!

